



TASK

Capstone Project – Full Stack With React and Express

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Introduction

Great job on making it to this point. In this Capstone Project, you will build a full-stack web application using React and Express that interfaces with the [iTunes Search API](#). The application should allow users to search for content within the iTunes Store and Apple Books and save a list of their favourites. This project will showcase your skills in both front- and back-end development, API integration, and basic documentation.

Instructions

This project involves building a web application that integrates a front end and a back end. The application will allow users to search for media using the iTunes Search API and perform basic interactions such as adding and removing favourites.



Practical task

Please consult the official documentation for an overview of the [iTunes Search API](#) to learn more about it. Then, carefully follow the instructions and ensure that your web application fulfils the following criteria.

Front end:

- Create an attractive and user-friendly interface using React components.
- Implement a search input where users can enter a search term and also select the type of media they want to search for (e.g., “movie”, “podcast”, “music”, “audiobook”, “short film”, “TV show”, “software”, “ebook”, or “all”).
- Implement a favourites list where users can add and remove items. The list of favourite content does not need to be remembered when the user exits your web application.
- All output results should be attractively displayed. Use a responsive framework like Bootstrap for a clean, mobile-friendly layout.

Back end:

- Use Node.js with the Express framework to build a server that delivers your React front end to users and processes API requests.
- Implement a route that communicates with the iTunes Search API to fetch music data based on user queries. Retrieve album information including album name, artist name, album cover image, and release date.
- Pass the API endpoints data to the front end as needed to display results to the user.
- Use JWT to authorise API requests and secure your API.

Take note:

- You do not have to authenticate users.
- You do not have to create a database for this application.
- No information about the users of your application, their favourites list, nor their history activity needs to be stored by your application after the user leaves the app.

Documentation:

- Create a **README.md** file that includes a description of your application, its purpose, and key features.
- Provide clear instructions on how a user can install and run your application locally.
- README files are written in Markdown, a plain-text format for project documentation. Use [Markdown](#) syntax for headers, lists, links, and images. Create one in a code editor like Visual Studio Code and save it with a **.md** extension.
- **Optional:** Create [basic wireframes](#) using [draw.io](#) to demonstrate how the user interface of your application will work. Export the wireframes as a PDF and save it in your task folder as **wireframes.pdf**.

Guidelines:

- The file structure of the project should be well-organised and easy to understand and use.

- Your code should be well-documented with appropriate comments. The code should also be easy to read, adhering to [Google's JavaScript Style Guide](#) about indentation, meaningful variables, component names, etc.
- You should ensure that your application has been appropriately tested and is free from errors or bugs. It's recommended to try using unit testing to help with this process.

Optional:

- Deploy your application on Render and add the link to your application in the **README.md** file of your project.
- Please note that you can only deploy static apps, hosted on GitHub, to Render. In order to deploy a dynamic application, like this Capstone Project app, you require a paid Render membership.

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.



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